SPECTRUM FRONT

FPS Gameplay Scripts — Missions 1–3

Shared interaction language

These three missions establish the controls and interaction grammar used throughout the game.

Core controls

Input	Action
WASD	Move
Mouse	Look
Shift	Sprint
C or Control	Crouch
E	Interact
Q	Equipment wheel
F	Switch between weapon and last instrument
Left mouse	Use instrument or fire
Right mouse	Precision mode or aim down sights
Mouse wheel	Adjust active instrument setting
R	Retune, recalibrate, or reload
M	Tactical map
Tab	Evidence notebook
H	Contextual hint
Escape	Pause and release pointer lock

Objective language

Objectives must always describe a physical action.

Do not display:

> Analyze the signal.

Display:

> Tune the analyzer between 430 and 470 MHz and record the signal that repeats every four seconds.

Evidence system

Every meaningful measurement creates an evidence card.

An evidence card contains:

- instrument
- player position
- time
- measured value
- screenshot-like miniature of the instrument display
- plain-language observation
- confidence
- mission relevance

Example:

> **Signal C**
> 452.8 MHz
> Burst every 4.0 seconds
> Bandwidth approximately 24 kHz
> Weak but repeatable
> Recorded at the forward command tent

Evidence cards are automatically attached to later decisions.

Lesson-card behavior

A lesson card appears only after the player has experienced the relevant phenomenon.

Each card:

- lasts no more than 12 seconds
- contains one concept
- uses the player's actual measurement
- does not freeze the game unless safety requires it
- can be reopened from the notebook

Hint ladder

Pressing H produces progressively more specific help.

Hint 1: Strategic

Explains what type of evidence is missing.

Hint 2: Instrument

Suggests which tool or setting to use.

Hint 3: Procedural

Gives exact steps but reduces the mission’s independence score.

MISSION 1

FIND THE MISSING EVACUATION SIGNAL

Mission identity

Primary concept: Frequency and tuning
Secondary concepts: Repetition interval, observation time, signal versus noise
Estimated duration: 12–18 minutes
Location: Forward operating base and casualty collection point
Player role: Newly assigned spectrum specialist
Primary instrument: Handheld spectrum analyzer

Operational stakes

A casualty evacuation helicopter is approaching through poor visibility.

The landing-zone beacon is expected to transmit, but the forward command post cannot receive it. The aircraft will abort if the pickup zone cannot be authenticated before reaching its decision point.

The player must locate the beacon’s signal and configure the authorized receiver before the extraction window closes.

The player is not searching for a definition of frequency. The player is trying to prevent an evacuation from being aborted.

Starting state

Environment

The player begins outside a canvas command tent at dawn.

Visible elements:

- idling armored ambulance
- medics treating two casualties
- radio mast
- command vehicle
- generator
- equipment cases
- helicopter audible but not yet visible
- blowing dust
- intermittent distant impacts

UI

Upper-left mission panel:

> **MISSION 1: FIND THE EVACUATION BEACON**
> Aircraft decision point in 06:00
> Report to the command tent.

Top-center countdown:

> EXTRACTION WINDOW 06:00

The countdown should not begin until the opening dialogue ends.

Opening sequence

Trigger

Player enters the command tent.

Characters

- Sergeant Vale, communications lead
- Medic Torres
- Helicopter controller over radio

Dialogue

Sergeant Vale:

> “The evacuation beacon should be calling every four seconds, but our receiver is deaf. The aircraft is already inbound.”

Medic Torres:

> “We have two critical patients. If they wave off, we wait another twenty minutes.”

Helicopter controller, distorted radio:

> “Specter One is approaching the decision point. Authentication required before final approach.”

Sergeant Vale:

> “Beacon should be somewhere between four-thirty and four-seventy megahertz. Exact channel was lost when the terminal reset.”

He opens the equipment case.

HUD objective

> Open the equipment case and take the spectrum analyzer.

Stage 1: Equip the analyzer

Player action

Press E at the open equipment case.

Instrument pickup animation

- Player lowers weapon.
- Gloved hands lift a rugged analyzer.
- Antenna unfolds.
- Screen boots with a brief diagnostic sweep.
- Device initially displays 390–410 MHz.
- No relevant signal is visible.

Analyzer screen

Required elements:

- current center frequency
- current span
- spectrum trace
- waterfall
- noise-floor estimate
- marker cursor
- scan-hold indicator
- record button prompt

Radio guidance

Sergeant Vale:

> “You’re looking at the wrong part of the spectrum. Use the tuning wheel. Bring the display into the four-thirty to four-seventy range.”

HUD objective

> Tune the analyzer into the 430–470 MHz search region.

Success trigger

Analyzer’s visible range overlaps at least 430–470 MHz.

Stage 2: Understand tuning through movement

The search region contains four signals.

Signal A

- 438.1 MHz
- strong
- frequent voice bursts
- nearby vehicle radio

Signal B

- 446.5 MHz
- continuous narrow carrier
- base telemetry clock

Signal C

- 452.8 MHz
- weak
- short burst every 4.0 seconds
- evacuation beacon

Signal D

- 466.4 MHz
- irregular digital bursts
- logistics terminal

The player should not initially receive labels.

Player action

The player turns the frequency control.

The analyzer display slides across the spectrum. Signals enter and leave the visible region.

Required behavioral lesson

The player must hold the analyzer on a region long enough to observe repetition.

Signal C should be invisible between bursts.

If the player scans too quickly, they miss it.

Sergeant Vale dialogue after rapid scanning

> “Slow down. A silent screen does not mean an empty channel. The beacon only speaks once every four seconds.”

HUD objective

> Hold each candidate region long enough to observe its timing.

Visual teaching

A small observation timer appears when the player stops tuning:

> Observing: 0.0 s

The timer fills to five seconds.

Stage 3: Save candidate signals

Player action

Aim the marker at a peak and hold left mouse for one second.

Instrument animation

- Marker locks to peak.
- Frequency readout enlarges.
- A circular recording indicator appears.
- Waterfall history freezes briefly into an evidence thumbnail.

Evidence cards

The player may record multiple signals.

Each receives a neutral name:

- Candidate A
- Candidate B
- Candidate C
- Candidate D

The objective requires at least two recorded candidates before the game allows final configuration.

HUD objective

> Record candidate signals and identify the one repeating every four seconds.

Correct evidence

Candidate C card:

> 452.8 MHz
> Short burst
> Repetition interval: 4.0 s
> Signal currently weak
> Pattern matches expected evacuation beacon timing

Lesson card trigger

Appears when Candidate C is recorded.

> **FREQUENCY**
> You moved the receiver to 452.8 MHz to observe this signal.
> Frequency identifies where a repeating electromagnetic signal lies within the spectrum.

The card minimizes automatically into the notebook.

Stage 4: Configure the receiver

Trigger

Player returns to the command receiver console after recording Candidate C.

Console interaction

The console has:

- six-digit frequency entry
- audio monitor
- authentication status
- signal-strength bar
- confirm button

Player action

Enter the measured frequency manually.

The game should not automatically fill it.

Correct input

452.800 MHz, with reasonable rounding accepted.

Incorrect input behaviors

Strong vehicle signal selected

Radio chatter plays, but no authentication code appears.

Console message:

> Signal received. Evacuation identifier absent.

Continuous telemetry carrier selected

Steady tone.

Console message:

> Carrier detected. Required burst timing absent.

Nearby but incorrect frequency

Noise or partial burst.

Console message:

> Insufficient signal lock.

Recovery

The player can reopen the notebook and inspect evidence cards without leaving the console.

Stage 5: Authenticate the landing zone

Correct-frequency result

The console receives a repeating digital identifier.

Display:

> LZ BEACON 7A
> AUTHENTICATION VALID
> RANGE CONFIDENCE: MODERATE

Dialogue

Sergeant Vale:

> “That’s it. Patch it to the aircraft.”

The player presses E on the transmit key.

Player character:

> “Specter One, beacon acquired. Landing zone Seven Alpha authenticated.”

Helicopter controller:

> “Seven Alpha confirmed. Continuing approach.”

The helicopter crosses overhead toward the landing zone.

The evacuation timer changes to:

> APPROACH COMMITTED

Medics begin loading casualties.

Mission completion debrief

A brief in-world debrief appears on the command screen.

> You did not choose the strongest signal.
> You tuned through the search region, observed long enough to identify a repetition pattern, and matched the signal to the mission requirement.

The game highlights:

- frequency used
- observation time
- number of false candidates recorded
- time remaining
- whether hints were used

Hint ladder

Hint 1

> The beacon’s identifying feature is its timing, not its signal strength.

Hint 2

> Use the analyzer between 430 and 470 MHz. Stop on each signal for at least five seconds.

Hint 3

> Tune near 453 MHz and watch for a short burst repeating every four seconds.

Failure and recovery states

Extraction timer reaches zero before authentication

The aircraft initially announces:

> “Unable to authenticate. Beginning wave-off.”

The mission does not hard-reset immediately.

Sergeant Vale says:

> "Keep working. If we recover the beacon quickly, we may get them on the second pass."

A new timer begins:

> SECOND PASS AVAILABLE IN 04:00

Mission score is reduced, but the player can still complete the lesson.

Player selects wrong signal repeatedly

After three failed console entries:

Sergeant Vale:

> "Go back to the requirement. We know how often the beacon transmits. Use that to separate it from the others."

Player never observes long enough

The analyzer flags:

> Observation window too short to estimate repetition.

Scoring

Total: 1,000 points

Category	Points
Correct beacon frequency	300
Timing pattern correctly recorded	200
Extraction completed on first pass	200
No more than one incorrect console entry	100
Minimal hint use	100
Evidence notebook complete	100

Mastery condition

The player must correctly tune the receiver using measured frequency and timing.

Mission completion alone is not enough if the console is auto-filled through hints.

MISSION 2

KEEP THE AID STATION ALIVE

Mission identity

Primary concept: Bandwidth
Secondary concepts: Receiver filtering, adjacent-channel interference, A/B testing
Estimated duration: 18–25 minutes
Location: Field medical station and logistics vehicle park
Primary instruments: Spectrum analyzer and network diagnostic tablet

Operational stakes

A field aid station is receiving casualties after an attack.

Wireless patient monitors function normally until a logistics convoy enters the compound and activates its high-power radios.

The player must restore telemetry without shutting down logistics communications needed to coordinate ammunition and casualty transport.

The game should show that preserving one system by destroying another is not a successful solution.

Starting sequence

Environment

- several field beds
- medical monitors
- nurses and medics moving between patients
- medical receiver mounted near the tent entrance
- approaching convoy visible through dust
- logistics vehicles parking nearby
- radio antennas rising from vehicles

Opening event

The player arrives as the monitors function normally.

A medic gestures at the network tablet.

Medic Torres:

> "Everything is stable right now. The failures begin when the convoy checks in."

HUD objective

> Record a baseline for the medical telemetry network before the convoy transmits.

Stage 1: Record normal operation

Player action

Equip network tablet and point it toward the medical network node.

Tablet display

- patient devices: 6 connected
- packet loss: 1–2%
- median latency: 42 ms
- retransmissions: low
- medical channel center frequency
- receiver filter width
- signal-quality history

Player action

Hold left mouse to record baseline.

Evidence card

> Medical telemetry baseline
> Packet loss: 2%
> All six monitors updating
> Receiver filter width: 260 kHz
> No active outage

HUD objective update

> Wait for convoy radio check and record what changes.

Stage 2: Trigger the outage

Trigger

Player remains in medical area for at least five seconds after baseline.

Event

A logistics vehicle commander begins a radio check.

Logistics radio:

> “Convoy Two, radio check. All vehicles report.”

A strong nearby transmission begins.

Visible consequences

- patient displays freeze intermittently
- packet-loss graph rises
- one monitor sounds an alarm
- medics switch to manual checks
- telemetry status falls from green to red

Tablet data

- packet loss: 46%
- retransmissions: very high
- medical signal still present
- connection not completely lost

Medic dialogue

> “It happens exactly like this every time. We need both networks working before the next casualties arrive.”

Countdown

> NEXT CASUALTY ARRIVAL 07:00

Stage 3: Compare the signals

HUD objective

> Use the spectrum analyzer to measure the medical and logistics signals.

Player action

Equip analyzer.

The display automatically opens near the medical channel but does not identify either signal.

Spectrum behavior

The player sees:

- medical signal: moderate-power, approximately 180 kHz wide
- logistics signal: much stronger, approximately 240 kHz wide
- partial adjacent overlap at the medical receiver's current filter setting

Measurement interaction

The player selects the medical signal.

Two draggable bandwidth markers appear.

Player drags the left and right markers to the visible edges.

The instrument calculates:

> Occupied bandwidth: approximately 180 kHz

The player records it.

They then measure the logistics signal.

> Occupied bandwidth: approximately 240 kHz
> Signal power at medical receiver: very high

Lesson card trigger

When the player finishes the first bandwidth measurement:

> **BANDWIDTH**
> This telemetry signal occupies approximately 180 kHz.
> A signal occupies a range of frequencies, not one infinitely thin line.

Stage 4: Experiment with the receiver filter

Trigger

Both signal measurements recorded.

The authorized receiver-filter control unlocks on the network tablet.

Interface

A live spectrum appears with a translucent receiver window.

The player drags:

- left filter edge
- right filter edge

Packet loss updates in real time after a short stabilization delay.

Three teachable states

State A: Filter too narrow

The player clips the desired signal.

Visual:

- outer portions of medical trace dim red
- packet decoder reports missing components
- packet loss rises to 30–60%

Tablet message:

> Desired signal not fully contained within receiver passband.

Medic consequence:

A monitor freezes despite reduced logistics interference.

State B: Filter too wide

The full medical signal is admitted, but significant logistics energy also enters.

Packet loss remains high.

Tablet message:

Strong neighboring energy admitted by receiver filter.

State C: Filter correctly placed

The medical signal fits fully inside.

Most logistics energy is rejected.

Packet loss drops to approximately 12–15%, but does not reach mission threshold because the logistics signal remains excessively powerful and close.

Contextual dialogue

Medic Torres:

“Better, but not good enough. What’s still getting through?”

HUD objective

Improve medical telemetry below 5% packet loss without disabling logistics service.

Stage 5: Conduct an A/B test

The player should prove the relationship rather than assume it from proximity.

Player action

Approach logistics radio operator.

Press E:

Request five-second radio-silence test

NPC response

Logistics operator:

“Five seconds only. Starting on your mark.”

Player confirms.

Test result

When logistics transmitter turns off:

- medical packet loss falls to 2%
- all monitors resume
- neighboring spectrum disappears

When it turns back on:

- packet loss rises again

Evidence card

Controlled A/B test
Logistics transmitter off: medical packet loss 2%
Logistics transmitter on: medical packet loss 14% after filter correction
Evidence supports interference from logistics transmission

Teaching card

Controlled testing changes one factor while holding others as constant as possible. The medical network recovered when the logistics transmitter stopped.

Stage 6: Apply the least-disruptive remedy

The player gains access to a simplified frequency plan on the tablet.

Available actions

1. Move logistics radio 500 kHz upward.
2. Move medical telemetry to a new channel.
3. Reduce logistics power by 6 dB.
4. Increase medical power.
5. Disable logistics radio.
6. Move the logistics antenna twenty meters away.

The best complete solution may combine:

- move logistics channel upward
- reduce excessive logistics power modestly

Required interaction

The player physically walks to the logistics vehicle and changes the authorized radio setting.

A tuning animation shows the operator confirming the new channel.

Consequence simulation

Correct reassignment

- medical packet loss: 2–3%
- logistics connectivity: stable
- no conflict with other local systems

Reassign medical channel carelessly

May overlap an emergency voice network.

New radio call:

> “Command, emergency net is breaking up.”

Increase medical power

Some improvement, but the receiver remains vulnerable.

Shut down logistics network

Medical system recovers, but convoy loses movement coordination.

Mission not complete.

Stage 7: Verify the solution

The mission requires three explicit checks.

Check 1

Record medical packet loss below 5%.

Check 2

Use logistics radio to complete a test call.

Check 3

Open local frequency plan and verify no new conflict warning.

Final dialogue

Medic Torres:

> “All monitors are updating.”

Logistics operator:

> “Convoy net is clean.”

Sergeant Vale:

> “That’s a fix. Not because the display turned green—because both systems still work.”

Mission debrief

The after-action display reconstructs the evidence chain:

1. Medical system normal before logistics transmission.
2. Packet loss rose during logistics activity.
3. Signals occupied neighboring frequency ranges.
4. Receiver filter admitted part of logistics transmission.
5. Controlled transmitter-off test restored telemetry.
6. Reassignment restored both systems.
7. Verification showed no new conflicts.

The debrief explicitly connects actions.

Hint ladder

Hint 1

> Compare what changes when the convoy begins transmitting.

Hint 2

> Measure the full width of both signals, not only their center frequencies.

Hint 3

> Narrow the medical receiver filter until it contains all of the medical signal but excludes as much logistics energy as possible. Then perform an authorized transmitter-off test.

Failure and recovery states

Casualty timer reaches zero with telemetry still degraded

New patients arrive.

Medics continue manually.

The mission remains playable, but:

- medical-risk score decreases
- more urgent audio and visual pressure begins
- player receives fewer opportunities for experimentation

Player disables logistics network

The aid station improves, but the convoy route becomes uncertain.

Sergeant Vale says:

> “You transferred the failure to someone else. Restore logistics service and solve the conflict.”

Player creates a new channel conflict

A new incident appears.

The player must either undo the assignment or resolve the new conflict.

Scoring

Total: 1,200 points

Category	Points
Baseline recorded	100
Both bandwidths measured accurately	200
Filter adjusted without clipping desired signal	200
Controlled A/B test completed	200
Both networks preserved	250
No new conflict created	150
Casualties arrive after restoration	100

Mastery condition

The player must demonstrate that bandwidth affects both desired-signal preservation and admitted interference.

Simply selecting a correct channel through a hint does not satisfy mastery.

MISSION 3

CHOOSE A LINK BEFORE THE UNIT IS CUT OFF

Mission identity

Primary concept: The electromagnetic spectrum as a finite operational resource
Secondary concepts: Frequency bands, bandwidth demand, propagation tradeoffs, spectrum planning
Estimated duration: 22–30 minutes
Location: Command-planning shelter followed by an industrial field deployment
Primary tools: Physical spectrum table, network tablet, terrain simulator

Operational stakes

A platoon is about to enter an industrial district where the existing relay is damaged.

The unit needs:

- command voice
- position reports
- medical telemetry
- drone imagery

The communications plan must be completed before the platoon crosses into the district.

A poor plan may fail because:

- there is not enough bandwidth
- multiple users overlap
- terrain blocks the selected links
- a high-rate service is placed in an unsuitable band
- critical services are given insufficient priority

The player must create a functioning plan rather than answer questions about the names of spectrum bands.

Starting sequence

Environment

The player enters a darkened command shelter.

At the center is an illuminated horizontal spectrum table approximately three meters long.

The table shows a continuous frequency axis.

The room also contains:

- terrain-model table
- communications-equipment racks
- planning screens
- live countdown
- platoon leader and communications staff

Countdown

> PLATOON DEPARTURE 12:00

Dialogue

Captain Idris:

> “Once the platoon crosses the rail yard, the old relay will not cover them. They need voice, positions, medical telemetry, and drone imagery.”

Sergeant Vale:

> “We do not have a separate clean channel for everything. Build a plan that survives the terrain.”

HUD objective

> Review the four communications requirements at the spectrum table.

Stage 1: Learn the requirements through interaction

Four physical requirement blocks sit beside the spectrum table.

The player picks each up with E and examines it.

Command voice block

- bandwidth requirement: narrow-to-moderate
- mobility: high
- latency sensitivity: high
- range: long
- priority: critical

Position-report block

- bandwidth requirement: very narrow
- update interval: every ten seconds
- delay tolerance: moderate
- priority: high

Medical telemetry block

- bandwidth requirement: moderate
- reliability requirement: very high
- priority: critical
- traffic: intermittent with emergency bursts

Drone imagery block

- bandwidth requirement: wide
- line-of-sight sensitivity: high
- priority: mission-dependent
- data rate: high

Each physical block has width proportional to its spectrum requirement.

The drone block is visibly much wider.

Teaching through manipulation

The player can hold each block over different portions of the spectrum table.

The table previews:

- available bandwidth
- congestion
- equipment compatibility
- rough propagation tendency
- antenna requirement

No answer is given automatically.

Stage 2: Place the position-report service

This is the simplest first placement.

Objective

> Place the low-rate position-report service into a compatible clear region.

The player drags the narrow block onto the table.

Several placements work.

Immediate simulation

Tiny location markers begin updating on the terrain display.

Teaching card

> Different services occupy different amounts of spectrum. Low-rate position reports require much less bandwidth than live imagery.

Stage 3: Place command voice

Available tradeoffs

Lower-frequency region

- better obstruction tolerance in the simplified simulation
- longer-range performance
- more crowded
- larger antenna requirement

Higher-frequency region

- more open spectrum
- smaller antenna
- weaker performance behind dense structures

Player action

Place the command voice block.

Test

Press “Run voice coverage test.”

A terrain overlay appears.

The player hears simulated radio calls from several platoon positions.

Poor high-frequency placement

Voice becomes intermittent behind the concrete processing plant.

Appropriate placement

Voice remains usable through most of the route.

Teaching card

> Different portions of the spectrum interact differently with antennas, terrain, and available channel width. Frequency choice affects more than the number entered into a radio.

Stage 4: Protect medical telemetry

The medical block requires:

- no direct overlap
- high priority
- moderate bandwidth
- low packet loss

The player places it.

If placed adjacent to an excessively powerful service, a caution appears only after running the simulation.

Possible outcome:

> Medical receiver overload during command transmissions.

The player must revise separation, power, or antenna arrangement.

Objective

> Preserve medical telemetry during simultaneous voice traffic.

The player runs a simulated emergency burst.

Success requires:

- medical packets delivered

- voice remains active
- no collision threshold exceeded

Stage 5: Place drone imagery

The drone block is too wide for several lower-frequency openings.

Player attempt outcomes

Force block into narrow gap

The block visually compresses, but the drone feed preview becomes:

- low resolution
- delayed
- frozen
- eventually disconnected

Tablet warning:

> Required data rate exceeds available channel capacity.

Use wider high-frequency region

The video feed works near line of sight but fails behind the warehouse.

Captain Idris dialogue

> “The channel carries the data. The terrain does not.”

New objective

> Add a relay that preserves the drone link through the industrial district.

Stage 6: Place a physical relay

The player moves to the terrain table.

Three candidate sites appear:

1. water tower
2. warehouse roof
3. low hill behind the command post

Player interaction

- Pick up relay model.
- Place at candidate site.
- Run terrain test.
- Watch coverage and link margin.

Candidate outcomes

Water tower

- excellent line of sight
- broad detectable footprint
- exposed position

Warehouse roof

- sufficient view into district
- moderate footprint
- vulnerable to nearby industrial noise

Low hill

- safe
- poor view into the deepest part of district

The best choice depends on current difficulty and scenario parameters, but the warehouse roof should usually offer the balanced solution.

Physical deployment

After selecting a plan, the player travels outside to the actual relay site.

They climb stairs or a ladder and install the relay.

This prevents the planning table from feeling detached from the battlefield.

Stage 7: Run the full operation

The player returns to the command shelter and presses:

> EXECUTE NETWORK TEST

The simulated platoon begins moving.

Live displays

- voice connection
- position updates
- medical packets
- drone video
- channel occupancy
- interference alerts
- relay status

Possible failure patterns

Spectrum overlap

Two service blocks flash simultaneously.

Real consequence:

- voice breaks up
- telemetry retransmits

Insufficient bandwidth

Drone feed freezes.

Terrain failure

Coverage disappears at specific map locations.

Poor priority configuration

Drone video consumes capacity during a medical emergency burst.

Player intervention

The player may pause the test once and revise:

- frequency placement
- bandwidth allocation
- transmit power
- schedule
- relay position
- service priority

Stage 8: High-stakes live inject

After the network passes the basic test, a simulated casualty occurs.

Event

A platoon member is injured while the drone is streaming video.

Medical telemetry sends an emergency burst.

Required behavior

The network must prioritize medical data.

If the player configured priorities correctly:

- medical burst passes immediately
- drone feed briefly reduces quality
- voice remains available

If configured poorly:

- medical data queues behind video
- medics receive delayed information

Captain Idris dialogue on success

> “Medical data came through. Video degraded, but the unit kept what mattered.”

This teaches spectrum management as mission prioritization rather than only geometric placement.

Mission completion sequence

The platoon crosses the district.

Displays show:

- voice available
- position reporting active
- medical channel protected
- drone feed routed through relay

Dialogue

Captain Idris:

> "The network is holding. Release the platoon."

The gate opens and the unit departs.

Teaching card

> **THE ELECTROMAGNETIC SPECTRUM**
> Every wireless system occupies part of a shared range of frequencies.
> Successful planning also depends on time, location, bandwidth, equipment, terrain, and mission priority.

Hint ladder

Hint 1

> Start with the services that need the least spectrum, then protect the critical services.

Hint 2

> The drone feed needs the widest clean region. Test terrain before committing to its band.

Hint 3

> Place position reports in a narrow clear channel, keep medical traffic separated and high priority, use a lower-frequency option for command voice, and support the higher-frequency drone link with a relay.

Failure and recovery states

Platoon departure timer expires before plan completion

Captain Idris delays departure by three minutes.

Operational score decreases.

The player receives one guided-planning overlay.

Network test fails repeatedly

After three failed full tests, the game highlights the first causal failure only.

Example:

> Drone service requires more bandwidth than currently assigned.

It does not solve every remaining problem.

Player maximizes every service

The table displays:

> Total requirement exceeds available spectrum.

The player must reduce:

- video quality
- update frequency
- simultaneous operation
- unnecessary redundancy

Player disables drone imagery entirely

The network technically works, but the platoon loses reconnaissance support.

Mission can complete at reduced operational score only if the drone requirement was noncritical in the scenario variant.

Scoring

Total: 1,500 points

Category	Points
All four services assigned	200
No destructive frequency overlap	250
Drone bandwidth requirement satisfied	200
Terrain-tested relay placed	250
Medical traffic correctly prioritized	250
Full network survives movement test	250
Plan completed before departure	100

Mastery condition

The player must demonstrate that:

- systems occupy different frequency widths
- the spectrum is shared and finite

- frequency choices affect propagation and equipment
- bandwidth, terrain, timing, and priority all shape a workable plan

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Cross-mission progression

These missions deliberately build on one another.

Mission 1 teaches

> A signal has a location in frequency.

Mission 2 adds

> A signal also occupies a width around that location.

Mission 3 expands

> Many systems must share the larger electromagnetic spectrum, and their placement must account for bandwidth, terrain, timing, and priority.

The player therefore develops a connected mental model:

> Find the signal
> → understand its occupied width
> → place multiple signals into a functioning operational environment

Reusable implementation rules established by Missions 1–3

1. The instrument display must be readable

The spectrum analyzer should not be decorative.

Players must be able to distinguish:

- center frequency
- visible span
- peaks
- occupied width
- timing
- noise floor

2. Every new term labels something already experienced

Frequency is named after tuning.

Bandwidth is named after clipping a signal.

Spectrum is named after competing services must be placed.

3. Consequences appear in the world

Incorrect settings change:

- helicopter decisions
- patient monitors
- convoy communication
- platoon network performance
- drone imagery
- medical response

4. Mission completion requires verification

Finding a frequency is not enough; the aircraft must authenticate.

Changing a filter is not enough; both networks must operate.

Creating a spectrum plan is not enough; it must survive the moving-unit simulation and emergency inject.

5. The next objective must be caused by the previous result

Example:

Filter adjustment improves telemetry but does not fully restore it
→ remaining interference suggests a second action
→ A/B test proves logistics involvement
→ channel controls unlock
→ reassignment becomes justified

The mission never advances solely because a scripted timer expires.